

(c) Indian Ocean

(d) Pacific Ocean

Uttarakhand P.C.S. (Pre) 2005

Ans. (a)

The point at which the Equator (0° latitude) and the Prime Meridian (0° longitude) intersect has no real significance but it is in the Gulf of Guinea in the Atlantic Ocean, so the correct option is (a).

4. Point of intersection of the Equator and the Prime Meridian is located at-

(a) Atlantic Ocean

(b) Ghana

(c) Morocco

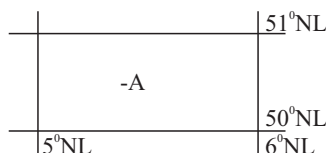
(d) Pacific Ocean

U.P. Lower Sub. (Pre) 2009

Ans. (a)

See the explanation of the above question.

5. Consider the geographical details given in the following figure :



The point marked by A in the above figure indicates the countries in :

(a) North America

(b) South America

(c) Europe

(d) Asia

I.A.S. (Pre) 1997

Ans. (c)

The given longitudes and latitudes in the question are approximately equivalent to the latitude and longitudes of the European country Belgium because this country falls near 5° East longitude and $50^\circ 30'$ North latitude. Therefore, the correct answer is option (c) Europe.

6. A plane starts flying from 30° northern latitude and 50° eastern longitude and lands at the opposite end of the earth, where will it land –

(a) 30° Northern latitude, 50° Western longitude

(b) 30° Southern latitude, 50° Western longitude

(c) 50° Northern latitude, 30° Western longitude

(d) 30° Southern latitude, 130° Western longitude

I.A.S. (Pre) 1994

Ans. (d)

At the opposite end of the earth, the plane will land at 30° southern latitude and 130° western longitude.

7. Among the following cities which one lies in the farthest North ?

(a) Beijing

(b) New Delhi

(c) New York

(d) Rome

U.P.P.C.S. (Mains) 2011

Ans. (d)

On the globe, the northernmost and southernmost positions of a place, city and country are decided by latitudes. The greater the angle of the latitude of a place in the southern or northern hemisphere the more southernmost or northernmost point, Position of the following cities are –

(City)	(Latitude)
Beijing (China)	$39^\circ 55' \text{ N}$
New Delhi (India)	$28^\circ 38' \text{ N}$
New York (U.S.A.)	$40^\circ 45' \text{ N}$
Rome (Italy)	$41^\circ 54' \text{ N}$

The Latitude of Rome is highest in comparison to the other cities. Therefore, Rome is the northernmost city among the given options.

8. The country which has the longest north-south (latitudinal) extension of its territory is :

(a) Russia

(b) Chile

(c) China

(d) Brazil

(e) None of the above/More than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (b)

Chile has the longest north-south (latitudinal) extension of its territory between 17° South and 56° South.

9. How many latitudes are there on the globe drawn at 1 degree interval?

(a) 180

(b) 178

(c) 179

(d) None of the above

69th B.P.S.C. (Pre), 2023

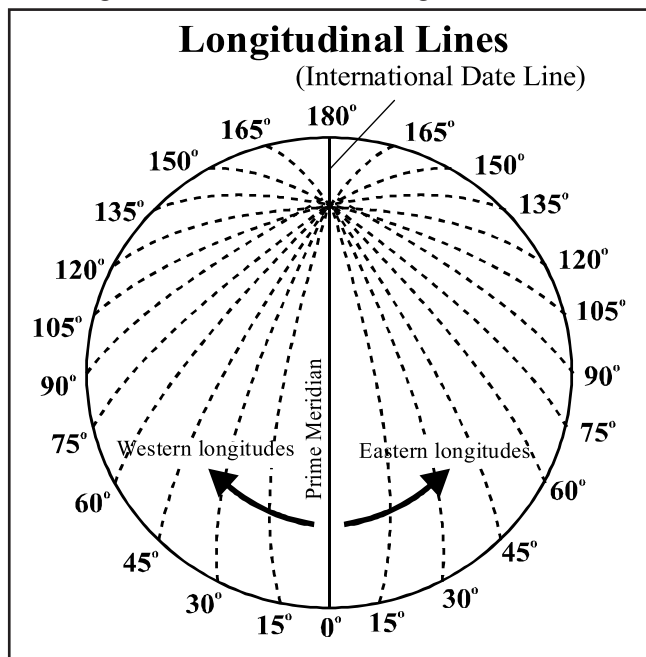
Ans. (c)

If parallels of latitude are drawn at an interval of one degree, there will be 89 parallels in the northern and the southern hemisphere each. The total number of parallels thus drawn, including the equator, will be 179.

ii. The Longitude

*Longitudes are imaginary lines joining the North and South Poles. They are also known as Meridians of Longitude or Lines of Longitude. *By convention, one of the longitudes was called the Prime Meridian. The longitudes of other places are measured as the angle east or west from the Prime Meridian, ranging from 0° at the Prime Meridian, to 180° eastward and 180° westward. In 1884, at the International Meridian Conference in Washington, D.C., 22 countries voted to adopt the **Greenwich meridian** as the Prime Meridian of the world. The 180° Meridian or

Antemeridian is the meridian 180° both east and west of the Prime Meridian. Thus, it is common to both east longitude and west longitude. Hence there are 360 longitudes.



*Since Earth makes one complete rotation of 360° in one day or **24 hours**, it passes through 15° in one hour or 1° in 4 minutes. The Earth rotates from west to east, so every 15° we go eastward local time is advanced by 1 hour. *The International Date Line, established in 1884 passes through the Mid-Pacific Ocean and roughly follows a 180° longitude North-South line on the Earth. It is located halfway around the world from the prime meridian. *When we cross the International Date Line from west to east, there is a decrease of one day and if you cross the line from east to west there is a gain of an extra day. This is because there is a difference of 24 hours between 180°E and 180°W longitude. 180°E and 180°W longitude are the same and written as only 180°. *The time at 180°E is 12 hours ahead and at 180° W it is 12 hours behind Greenwich time. A time zone is a region or area that observes the same standard time. The 360 longitudes have been divided into **24 time zones**. The world is divided longitudinally into time zones, with each hour difference roughly 15° apart. This means that every time zone has a difference of 1 hour and every longitude has a difference of 4 minutes. *The international standard time is determined by the prime meridian. *In the Northern Hemisphere the Prime Meridian passes through the **United Kingdom, France and Spain in Europe** and **Algeria, Mali, Burkina Faso, Togo and Ghana in Africa**. The only landmass crossed by the

Meridian in the Southern Hemisphere is **Antarctica**. *Apart from the United Kingdom there are a few more countries that despite being away from GMT (Greenwich Mean Time), use it for determining their standard time. These countries/areas are – Iceland, Ireland, Portugal, Western Sahara, Sao Tome Mauritania, Mali, Senegal, Guinea-Bissau, Guinea, Ghana, Liberia, Cote d'Ivoire, Burkina Faso, Gambia, Saint Helena etc. The time difference between Indian Standard Time (82°30' E longitude) and GMT is **5.30 hours**. ***Katchal**, a tiny island near Nicobar Island was virtually unknown until the Royal Greenwich Laboratory declared that it would be the first inhabited place on Earth to catch the rays of the first sunrise of the millennium. The **Katchal** island is located at a distance of 1600 km from the mainland of India and 418 km from Port Blair port. The island is located at 7°56' North latitude and 93° 24' East longitude.

***International Date Line (IDL)** is an imaginary line of demarcation on the surface of the Earth that runs from the North Pole to the South Pole and demarcates the change of one calendar day to the next. *IDL is roughly based on the meridian of 180° longitude, roughly down the middle of the Pacific Ocean and halfway around the world from the Greenwich Meridian. The IDL is deviated east or west away from the meridian eight places to prevent any anomalies regarding the date at a particular place.

*The difference between the International Date Line (180° longitude) and Indian Standard Time (82°30' East) is 97°30' longitude. International Date Line passes through the Bering Strait (Pacific Ocean).

1. The time at Cairo is 2 hours ahead of Greenwich. Hence, it is located at-
 (a) 30° W longitude (b) 30° E longitude
 (c) 28° E longitude (d) 28° W longitude

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (b)

Cairo (Egypt) is located at 30° 3' N latitude and 31° 14' E longitude. The time of Cairo is 2 hours ahead of Greenwich. Therefore its position approximately falls near the 30° East longitude. The Earth being spherical in shape rotates on its axis by 360° in 24 hours. Therefore, its movement for 1° of longitude takes the time of 4 minutes. So, it can be easily calculated that in 120 minutes (2 hours) the angle for longitude will be 30°. Therefore, option (b) should be the correct answer.

2. When it is noon along 82° 30' E longitude along what degree of longitude it will be 6.30 a.m.?
 (a) At 165° East (b) At 67° 30' West

- (c) At 0° East or West (d) At 82° 30' West

I.A.S. (Pre) 1994

Ans. (c)

When there is a midday at 82°30' E longitude then 6:30 in the morning means there is a difference of 5:30 hrs in time. As in one hour the change in longitude is 15° therefore 82° 30' difference in 5:30 hours is $82^{\circ} 30' - 82^{\circ} 30' = 0^{\circ}$ latitude.

- 3. If the difference of longitudes for the two places is 90°, the time interval between these places will be –**

- (a) 3 hr (b) 6 hr
(c) 9 hr (d) 12 hr

M.P.P.C.S. (Pre) 1992

Ans. (b)

Generally, 1° longitude is equal to a difference of 4 minutes. 90° longitude means $90 \times 4 = 360$ minutes or 6 hours. Hence (b) is the correct answer.

- 4. When, there is mid-day in the Greenwich, local time of a place is 5 O'clock evening. At which of the following longitude (meridians) that place will be located –**

- (a) 75° E (b) 75° W
(c) 150° E (d) 150° E

U.P.P.C.S. (Pre) 2008

Ans. (a)

According to the question, when it is mid-day at Greenwich there is 5 O'clock somewhere in the evening. This means the time of that place is ahead of Greenwich time. Hence, that place is located to the east of Greenwich. The Difference in time is 5 hours. Therefore,

1 hour = 60 minutes

5 hours = $5 \times 60 = 300$ minutes

∴ 4 minutes = 1°

$$300 \text{ minutes} = \frac{300 \times 1}{4} \Rightarrow 75^{\circ}$$

As this place is located to the east of the Greenwich line hence this place is located at 75° E.

- 5. Which one of the following longitudes along with the Prime Meridian forms a great circle on the globe.**

- (a) 0° (b) 90° E
(c) 90° W (d) 180°

U.P.P.C.S. (Pre) 2000

Ans. (d)

The Prime Meridian is the zero degree line of longitude that passes near London. The Prime Meridian was accepted as Zero degrees longitude at an international conference in 1884. It is also known as the Greenwich meridian. Just opposite to 0° longitude there is an imaginary line located at about 180° E or W as the International Date Line which forms a great circle with the Prime Meridian along the globe.

- 6. In the New Millennium, the first ray of the Sun was seen in which of the following Meridians of India –**

- (a) 2° 30' W (b) 82° 30' E
(c) 92° 30' W (d) 93° 30' E

U.P.P.C.S. (Pre) 2001

Ans. (d)

According to the Royal Greenwich-Laboratory's declaration the first ray of the sunrise, in India on the occasion of the New Millennium was seen upon Katchal Island near Andaman Nicobar Island. Katchal is situated nearly 1600 km away from the mainland of India and at a distance of 418 km from Port Blair Port and between the 7°56' North latitude and 93° 24' East Meridian. Therefore, the nearest option will be (d).

- 7. The basis of deciding the standard time of any place is–**

- (a) Longitude
(b) Latitude
(c) International Date Line
(d) Prime Meridian

43rd B.P.S.C. (Pre) 1999

Ans. (d)

Prime Meridian line is also known as the International Meridian or Greenwich Meridian. This line passes through England's Royal Greenwich Observatory. Line situated at zero degrees longitude determines the International Standard Time. 180° longitude is called the International Date Line. The Prime Meridian (0° longitude) passes through the following countries – (1) United Kingdom, (2) France, (3) Spain, (4) Algeria, (5) Mali, (6) Burkina-Faso, (7) Togo, (8) Ghana.

- 8. The Prime Meridian does not pass through.**

- (a) Algeria (b) France
(c) Niger (d) Spain

U.P. U.D.A./L.D.A. (Mains) 2010

Ans. (c)

See the explanation of the above question.

9. The Prime Meridian passes through

1. Algeria
2. France
3. Nigeria
4. Portugal

Use the code below to select the correct answer.

Code :

- | | |
|-------------|-------------|
| (a) 1 and 2 | (b) 2 and 3 |
| (c) 3 and 4 | (d) 1 and 3 |

U.P.P.C.S. (Spl) (Mains) 2008

Ans. (a)

See the explanation of the above question.

10. Time of which of the following places is equal to the time of GMT.

1. Accra
2. Dublin
3. Madrid
4. Lisbon

Use the code below to select the correct answer:

Code :

- | | |
|----------------|----------------|
| (a) 1, 2 and 3 | (b) 1, 3 and 4 |
| (c) 1, 2 and 4 | (d) 2, 3 and 4 |

I.A.S. (Pre) 1993

Ans. (c)

Greenwich Mean Time (GMT) is calculated from the Royal Greenwich Observatory (London) (at the 0 meridian). Some other countries that are away from it also use the time of GMT as given here-

Iceland	Sao Tome
Ireland	Liberia
Portugal	Cote d'Ivoire
Western Sahara	Burkina-Faso
Mauritania	Togo
Mali	Ghana
Senegal	Guinea
Guinea Bissau	Gambia
St. Helena	

This is notable that the Capital of Ghana, Ireland and Portugal respectively are Accra, Dublin and Lisbon which use the time of GMT, Madrid is the capital of Spain situated at 3° 43' Meridian at the West and its time is not equivalent to Standard Time (Time at Greenwich) or GMT.

11. Which one of the following cities does not have the same clock time as that of the other three Cities at any given instant?

- (a) London (UK)
- (b) Lisbon (Portugal)

- (c) Accra (Ghana)
- (d) Addis Ababa (Ethiopia)

I.A.S. (Pre) 2007

Ans. (d)

London (U.K.), Lisbon (Portugal) and Accra (Ghana) have similar time as all three use Greenwich time. Addis Ababa the capital of Ethiopia is located at 38° 45' E, whose time is 2 hours, 35 minutes, ahead of GMT.

Note - The Standard time of Ethiopia is 3 hours ahead of Greenwich time. So, Addis Ababa is the correct answer.

12. The standard time of the following countries is ahead of or behind Greenwich Mean Time depending on whether they are East or West of the longitude passing through Greenwich.

- | | |
|-----------|----------------|
| (1) Cuba | (2) Greece |
| (3) Iraq | (4) Costa Rica |
| (5) Japan | |

Which one of the following sequential orders gives the correct arrangement of the countries according to their standard time from ahead to behind GMT?

- | | |
|-------------------|-------------------|
| (a) 5, 3, 2, 1, 4 | (b) 2, 4, 1, 3, 5 |
| (c) 4, 1, 3, 2, 5 | (d) 3, 5, 4, 1, 2 |

I.A.S. (Pre) 1995

Ans. (a)

Greenwich line (0° longitude) passes through England. World standard time is measured from here. Time in Eastern Hemisphere is ahead of the Western Hemisphere. The standard time of the above mentioned countries is given below :

Japan = +9 hours

Iraq = + 3 hours

Greece = + 2 hours

Cuba = -5 hours

Costa Rica = -6 hours

Hence, the correct sequence of standard time ahead to behind GMT is 5, 3, 2, 1, 4 Japan, Iraq, Greece, Cuba and Costa Rica.

13. When there is noon at I.S.T. meridian people another place of the Earth is taking their 6 O'clock morning tea. The longitude of the place is :

- | | |
|---------------|-------------|
| (a) 17°30' E | (b) 7°30' W |
| (c) 172°30' E | (d) 90° W |

I.A.S. (Pre) 1998

Ans. (b)

The Earth makes one rotation (360°) in 24 hours, which means it passes through 15° in an hour from West to East. Every 15° we go Westward, local time becomes 1 hour behind the place, given in the question, which is 6:00 hours behind the I.S.T. ($82^\circ 30' \text{E}$). So it is situated to the west of the I.S.T. 6:00 hour difference is equal to 90° longitudes ($15 \times 6 = 90$). Thus, the place is situated at 90° west with respect to I.S.T. This gives the longitude of the place (with respect to Greenwich) as $7^\circ 30' \text{W}$ ($82^\circ 30' \text{E} - 90^\circ = 7^\circ 30' \text{W}$).

- 14. A telegram was sent from Greenwich at 12.00 Noon. The time for transmission was 12 minutes. It reached a town at 6.00 p.m. The longitude of the town will be :**
- (a) 97°E (b) 87°E
(c) 87°W (d) 97°W

Uttarakhand P.C.S. (Pre) 2012

Ans (b)

Earth rotates on its axis in 24 hours and therefore completes the angle of 360° in one rotation (i.e. 15° in an hour or 1° in 4 minutes). As the telegram sent from Greenwich at 12:00 noon reached another city at 6:00 evening and 12 minutes passed in messaging the telegram. The, total time = 6 hours – 12 minutes = 360 minutes – 12 minutes = 348 minutes. As we know for 60 minutes = 15° longitude
for 348 minutes = $15 \times 348/60 = 87^\circ$
This is now clear that place where the telegram reached at 6:00 evening will be at 87°East longitude.

- 15. International Date Line passes through –**

- (a) Africa (b) Asia
(c) Pacific ocean (d) Atlantic ocean

44th B.P.S.C. (Pre) 2000

Ans. (c)

International Date Line is an imaginary line on the Earth that separates two consecutive days on the calendar. This line passes through 180° opposite to Greenwich (England) from the Pacific Ocean. This line turns when any land area falls in the way of this line.

- 16. Statement (A) : Time interval at the date-line is 12 hours from the Greenwich.**

Reason (R) : Date-line is situated at 180° meridian.

Using following options choose the correct answer –

- (a) (A), (R) both are correct, and (R) is the correct explanation of (A).
(b) (A) and (R) both are correct, but (R) is not the correct explanation of (A).

(c) (A) is correct, but (R) is false.

(d) (R) is correct, but (A) is false.

U.P.P.C.S. (Pre) 1992

Ans. (a)

International Date Line at 180° longitude means that from the Greenwich (England). It is on the opposite side. Greenwich being situated at 0° Meridian the International Date Line has a difference of 180° longitude or $180^\circ \times 4 = 720$ minutes or 12 hours. Therefore, the 'Statement' and the 'Reason' both are correct and Reason (R) is the correct explanation of the Statement (S).

- 17. There is no minimum distance between two places on the globe :**

- (a) 45°N. Lat (b) 45°S. Lat
(c) Prime Meridian (d) International Date Line

U.P.P.C.S. (Pre) 1997

Ans. (d)

Except for the International Date Line (180° meridian), meridians at all other places follow the same path. The International Date Line has been turned at 8 places on the globe so that disputes about the date at the same place not occur. As the International Date Line has been turned on its way near the Alaska, therefore the minimum distance between two places on the globe at the International Date Line cannot exist.

- 18. If the earth's direction of rotation is reversed, what should be the IST when it is noon at the International Date Line?**

- (a) 06.30 hrs (b) 05.30 hrs
(c) 18.30 hrs (d) 17.30 hrs

I.A.S. (Pre) 1997

Ans. (a)

Time difference between International Date Line (IDL) and GMT is of 12 hours. Noon (12 pm) at IDL means midnight (12 am) at GMT. I.S.T. is GMT + 5:30 hours, but, in the question 'the Earth's direction of rotation is reversed'. So that I.S.T. becomes GMT-5:30 hours.
In this way, it should be 6:30 pm at I.S.T. (instead of 5:30am).

- 19. Which one of the following Straits is nearest to the International Date Line?**

- (a) Malacca Strait (b) Bering Strait

(c) Strait of Florida

(d) Strait of Gibraltar

I.A.S. (Pre) 2008

Ans. (b)

The Prime Meridian line or the Greenwich line is situated at 0° Meridian and on the 180° East or West side of it lies the International Date Line. So either from the East or from the West side of this line the place at the nearest meridian will be nearest to the Date Line. The meridian of straits given in the options is as follows –

Malacca - 99° 32' East Meridian

Bering Strait - 168° 23' West Meridian

Strait of Florida - 82° 23' West Meridian

Strait of Gibraltar - 5° 41' West Meridian

From the above data, it is clear that Bering Strait (168° 23' West Meridian) is the closest place to the International Date Line.

- 20. A ship is moving from the eastern side of Aleutian Islands towards Dutch Harbour. It crosses 180° meridian on 1st January 1999 at 23:30 hr. What time and date will be recorded by the captain of the ship in his diary after one-hour journey from the point of crossing of the meridian?**

- (a) January 1, 00:30 hr (b) January 2, 00:30 hr
(c) January 3, 00:30 hr (d) January 4, 00:30 hr

I.A.S. (Pre) 1999

Ans. (b)

Aleutian Island's eastern end lies at the west of 180° Meridian, (172° East longitude) while Dutch Harbour lies at the east side of 180° Meridian i.e at (166° 31' 55" West longitude). As the movement of the ship from West to East of the 180° Meridian, the date will lag behind by 1 day. So, at 23:30 hr. 1 January 1999 crossing 180 Meridian by rule the date should be 31 December 1998 and after 1 hour (23.30 + 60 = 2400 + 30) captain of the ship will record 00:30 hr. 1 January, 1999. But actually, the International Date Line does not completely follow the 180 Meridian to avoid segregating the places with equal political and economic entities (including Aleutian Islands and Dutch Harbor). In the context of this situation International Date Line is not crossed by the ship and the day will be same even after 1 hour the captain of the ship will record 00:30 hr. on 2 January, 1999.

- 21. Captain of a ship moving Westward records the local time of 10-00 hrs Monday at 90° W. If the speed of the ship is the same as the speed of the rotation of the earth, which of the following local time and day would be found by the captain at International date line:**

- (a) 04.00 Monday (b) 10.00 Monday

(c) 10.00 Tuesday

(d) 16.00 Tuesday

U.P.P.C.S. (Pre) 2000

Ans. (b)

Captain of the ship traveling westward recorded the local time 10:00 hours at 90° west Meridian, this means that the ship is near the North American continent. Since the Earth rotates from west to east on its axis (anticlockwise) and the ship is moving towards west with the speed equivalent to the Earth's rotation therefore, both are moving in opposite directions but there is no effect of Earth's rotation upon the movement of objects upon its surface. So, for the ship covering 90° meridian it requires the time of 6 hours and the ship will be behind 6 hours according to local time. Therefore, when reaching the International Date Line he will write the local time as the same 10:00 hrs Monday. However, when he crosses the International Date Line his local time will be 10:00 hours Tuesday.

- 22. Which path must be followed by a Ship to move from one place to another in the least time ?**

- (a) Sea or Ocean Current (b) Sea-breeze
(c) Meridian (d) Longitude

U.P.P.C.S. (Pre) 1992

Ans. (c)

For a ship to travel in the least possible time from one place to another, it must travel with respect to the Meridian. At the 180° meridian traveling from east to west, it gets one day less in this way while moving from west to east one day increases.

- 23. The local time of a place is 6:00 AM when the Greenwich Mean Time (GMT) is 3:00 AM, what is the longitude of the place ?**

- (a) 45° West (b) 45° East
(c) 120° East (d) 120° West

Uttarakhand P.C.S. (Pre) 2010

U.P.P.C.S. (Pre) 2006

Ans. (b)

Greenwich Mean Time (G.M.T.) is referred to as Coordinated Universal Time (U.T.C.). International time is calculated from Greenwich at the 0 Meridian, and the time eastward of it is while the time westward of it is behind (although dates are measured from the 180° meridian at the International Date Line which passes through the mid of the Pacific Ocean). Meridians situated 180° east of Greenwich are called East-meridians and meridians situated at 180° west are known as West-meridians. All the meridians in 360° of the globe are divided into 24 time